

Framework Design

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Chair of Decision Sciences & Systems

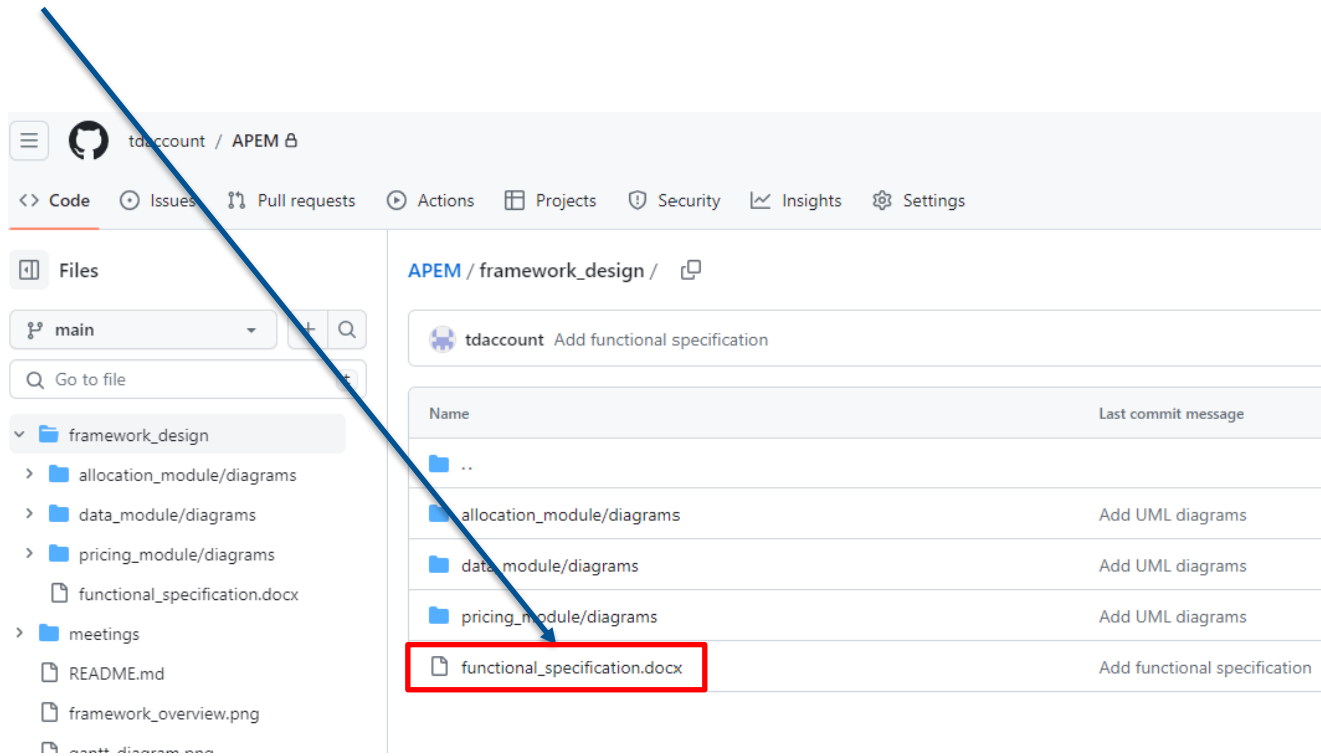
Technical University of Munich

APEM Workshop

February 8, 2024

Last Time: Use Cases, Functional Requirements

- The *functional specification* is included in our Github repository.

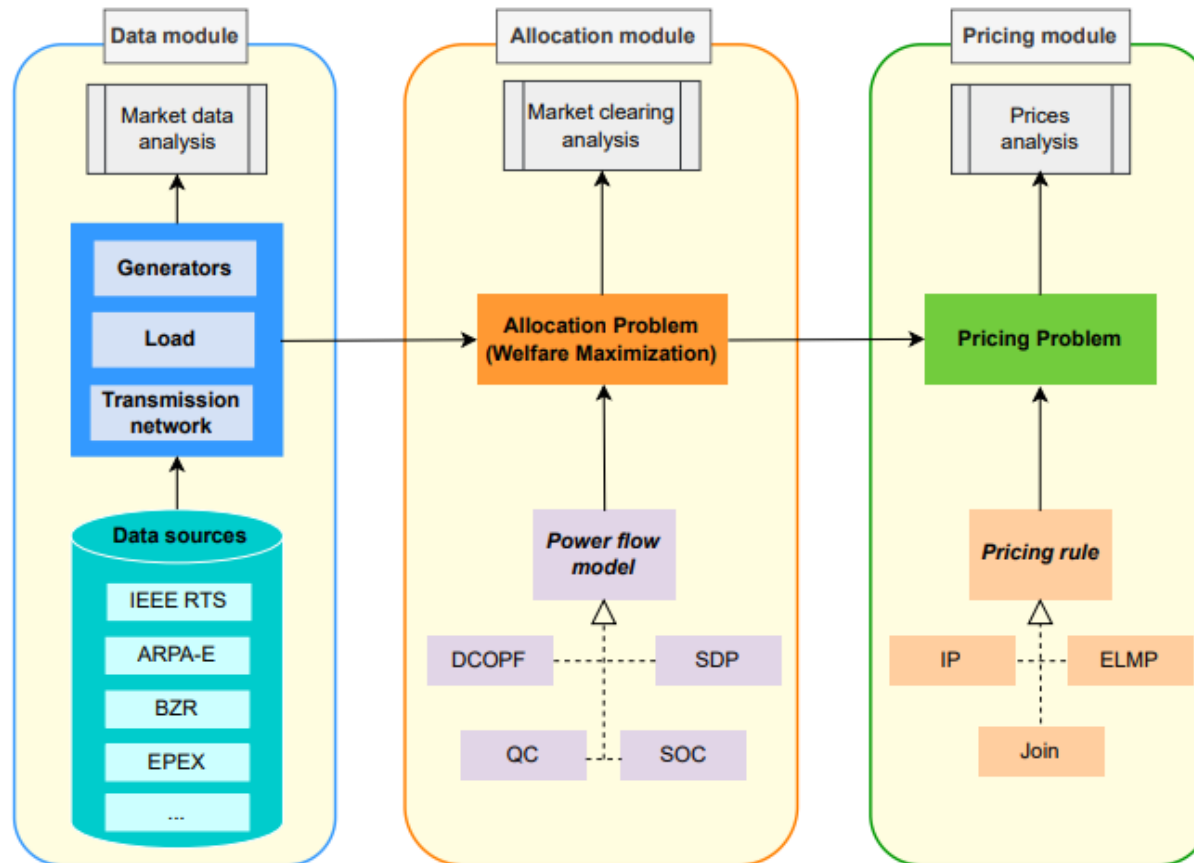


Today: Framework Design

Architecture:

- **Layered**
 - ✓ High coherence inside layers
 - ✓ Low coupling between layers
- **Call-and-return**
 - Scenario pipeline:
extract data → solve allocation → compute prices → analyze results
- **Object-oriented**

3 Modules (Layers)

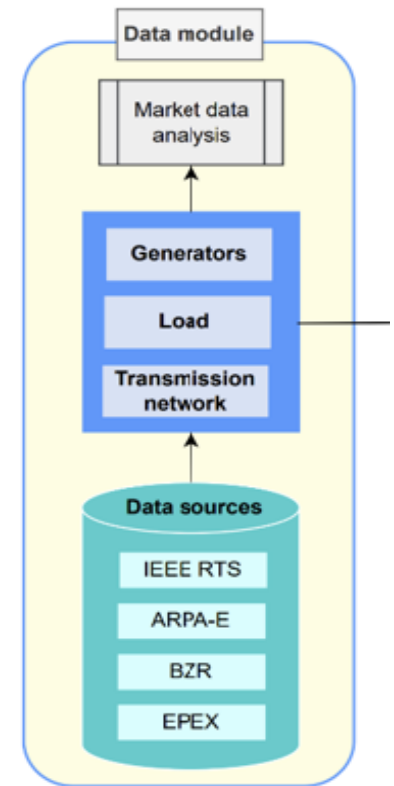


Goal: discuss the **static model** of each module

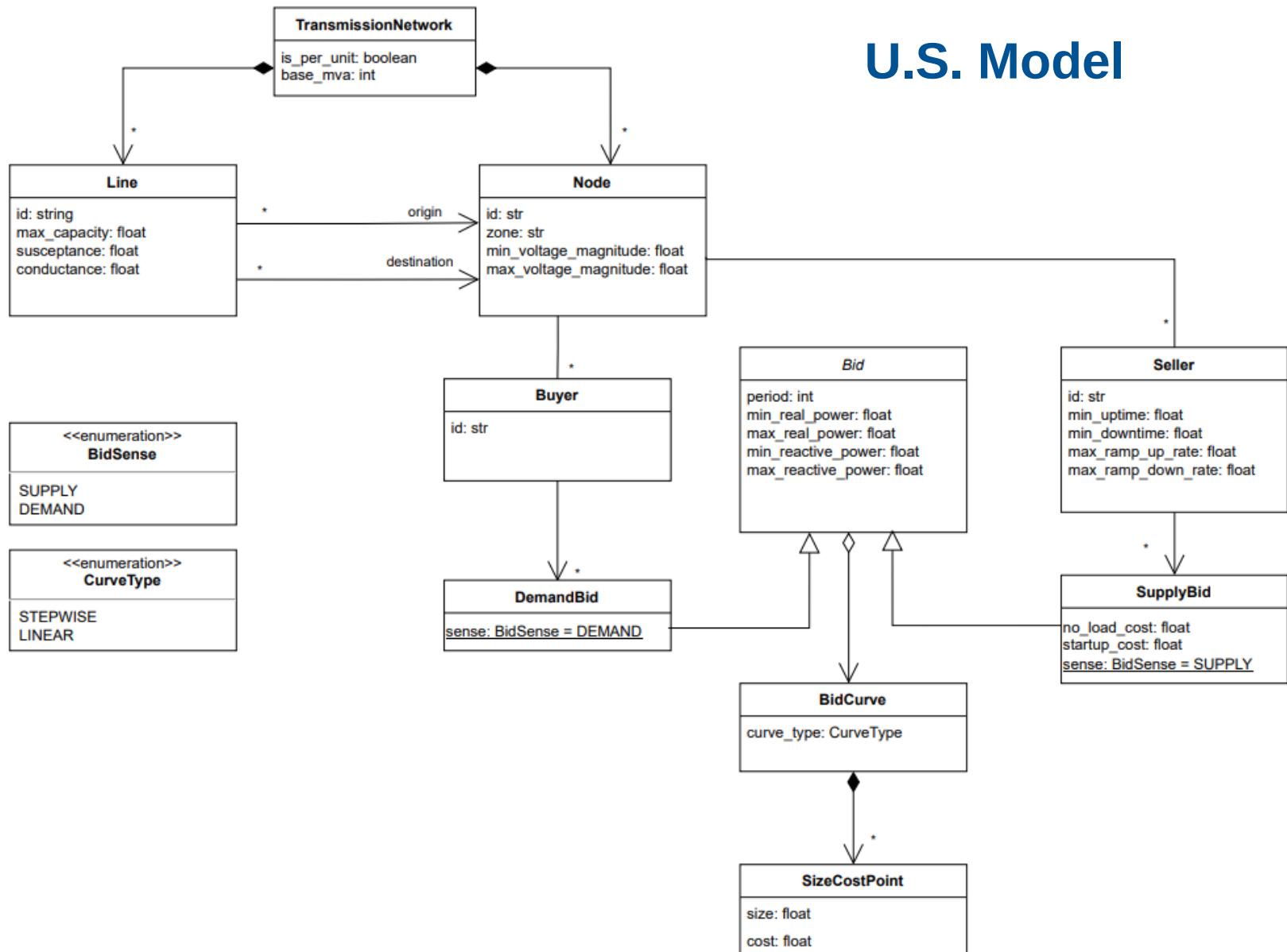
Data Module

U.S. and European models deal with different types of data.

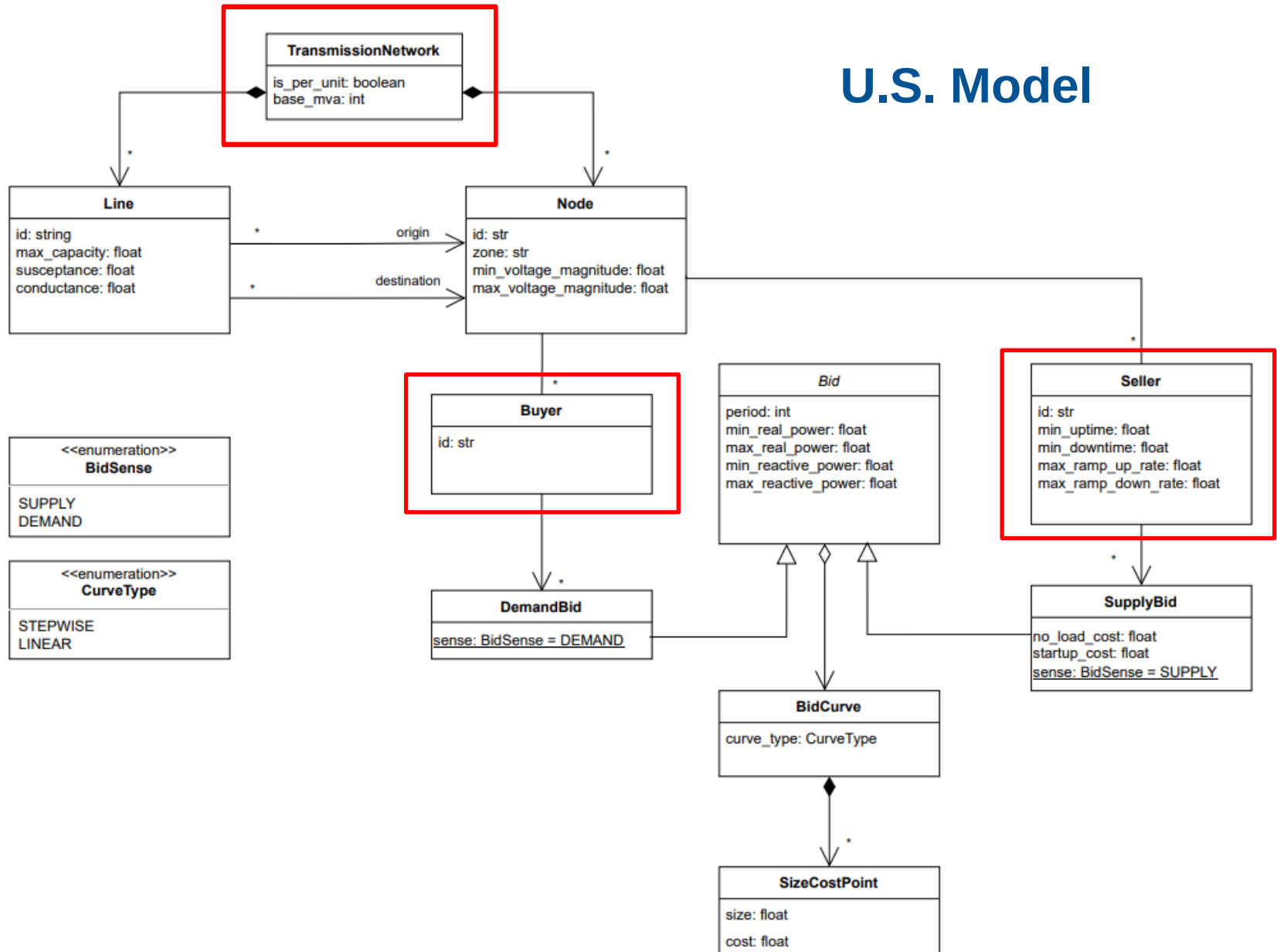
→ 2 data models



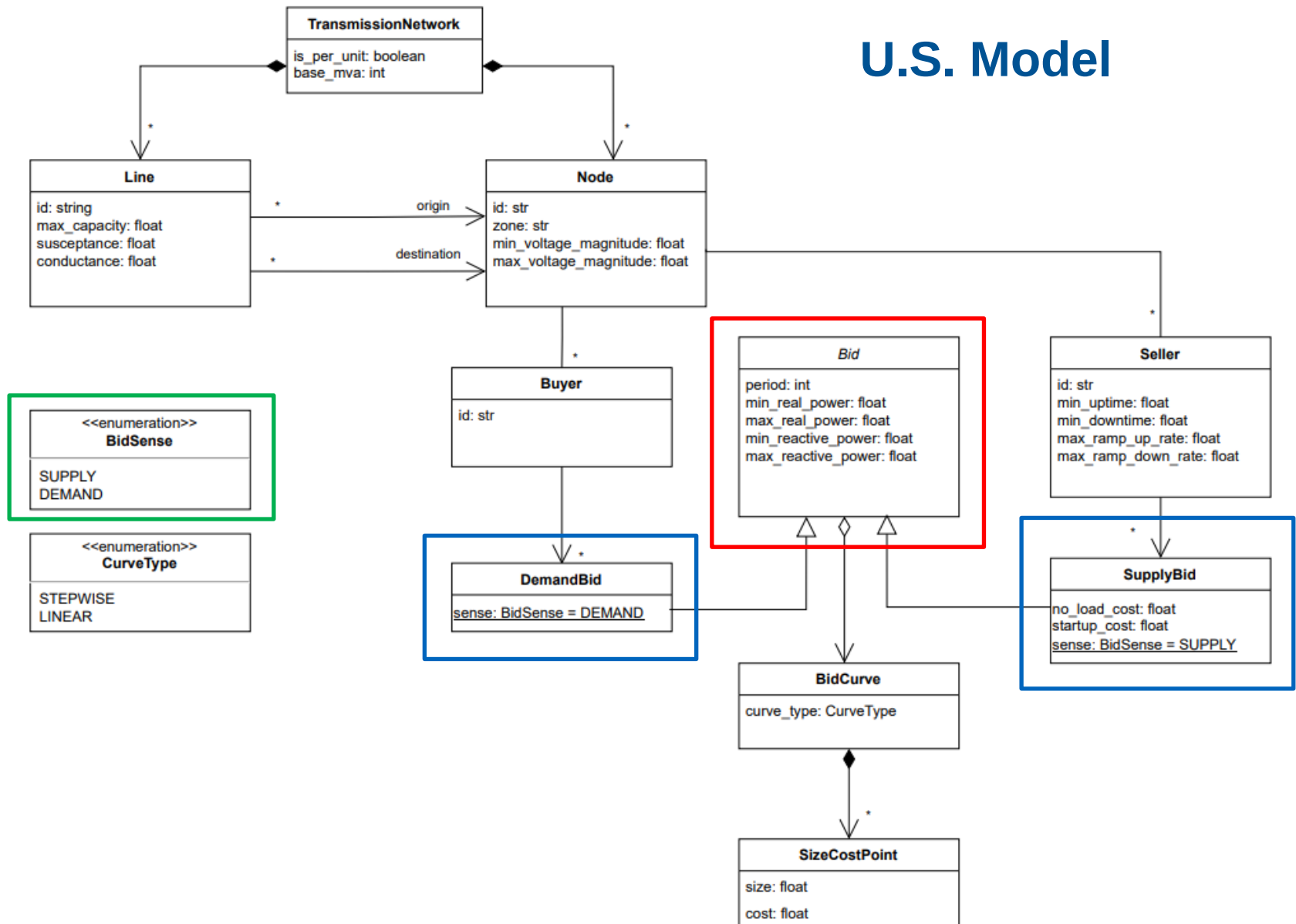
U.S. Model



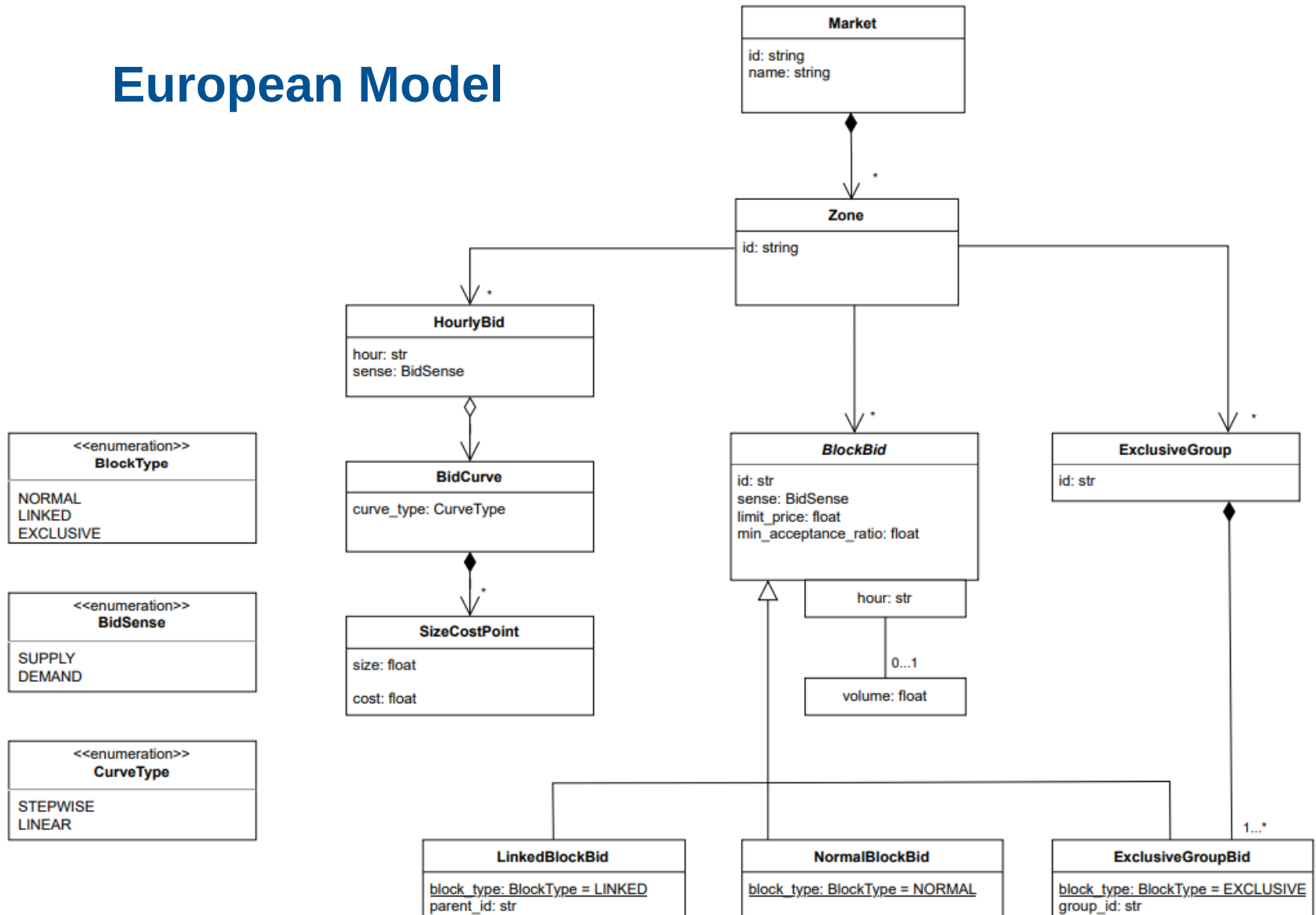
U.S. Model



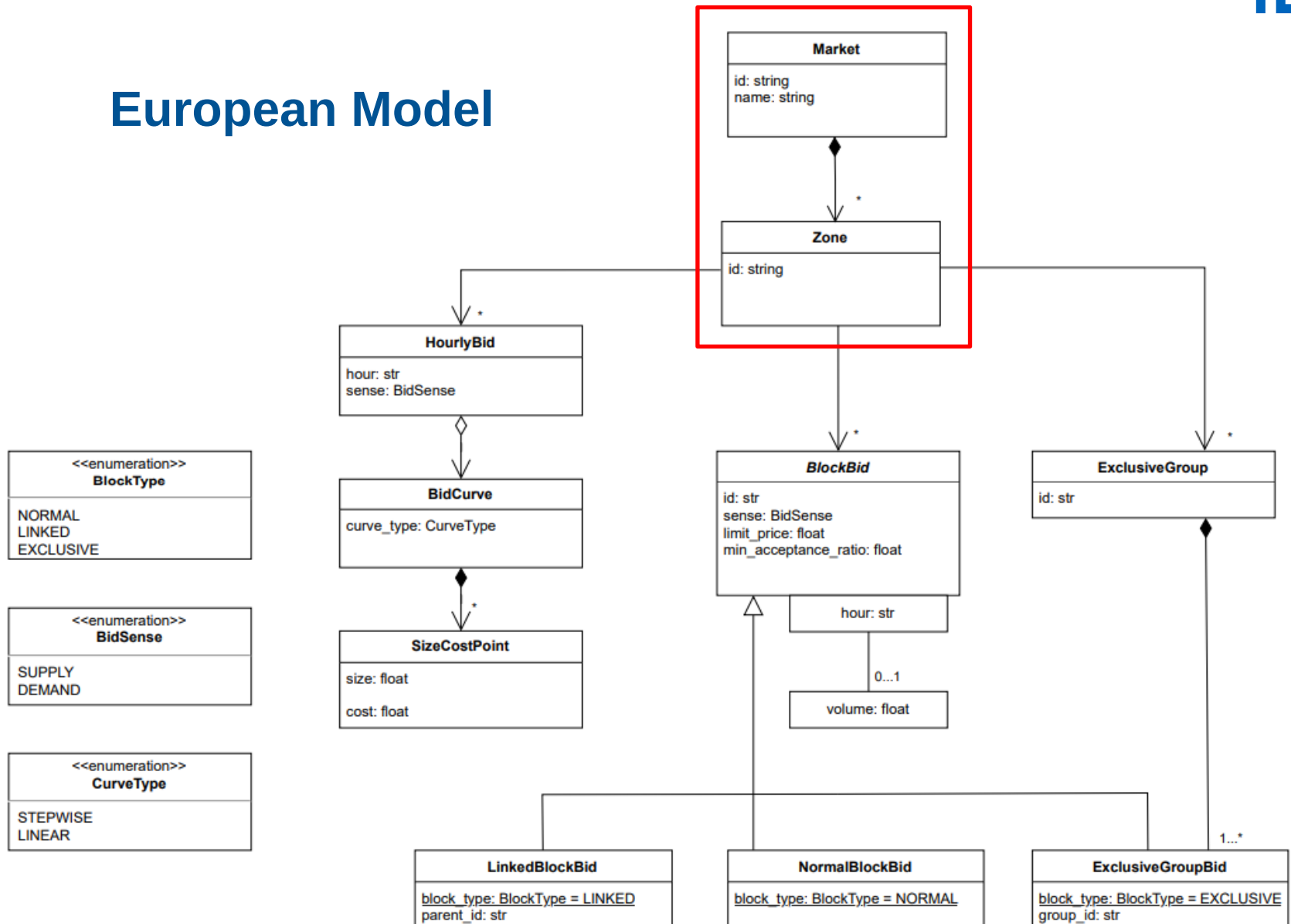
U.S. Model



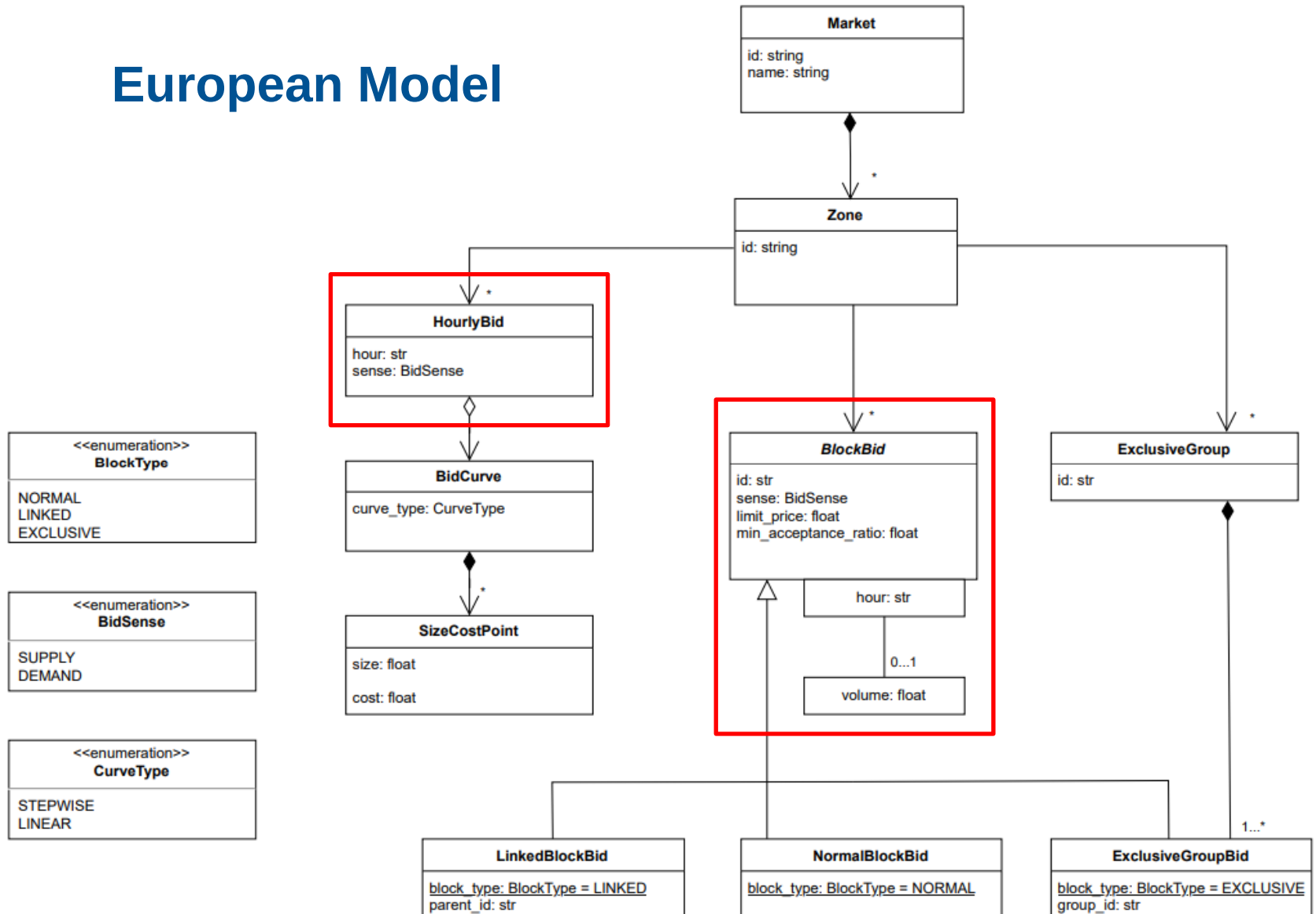
European Model



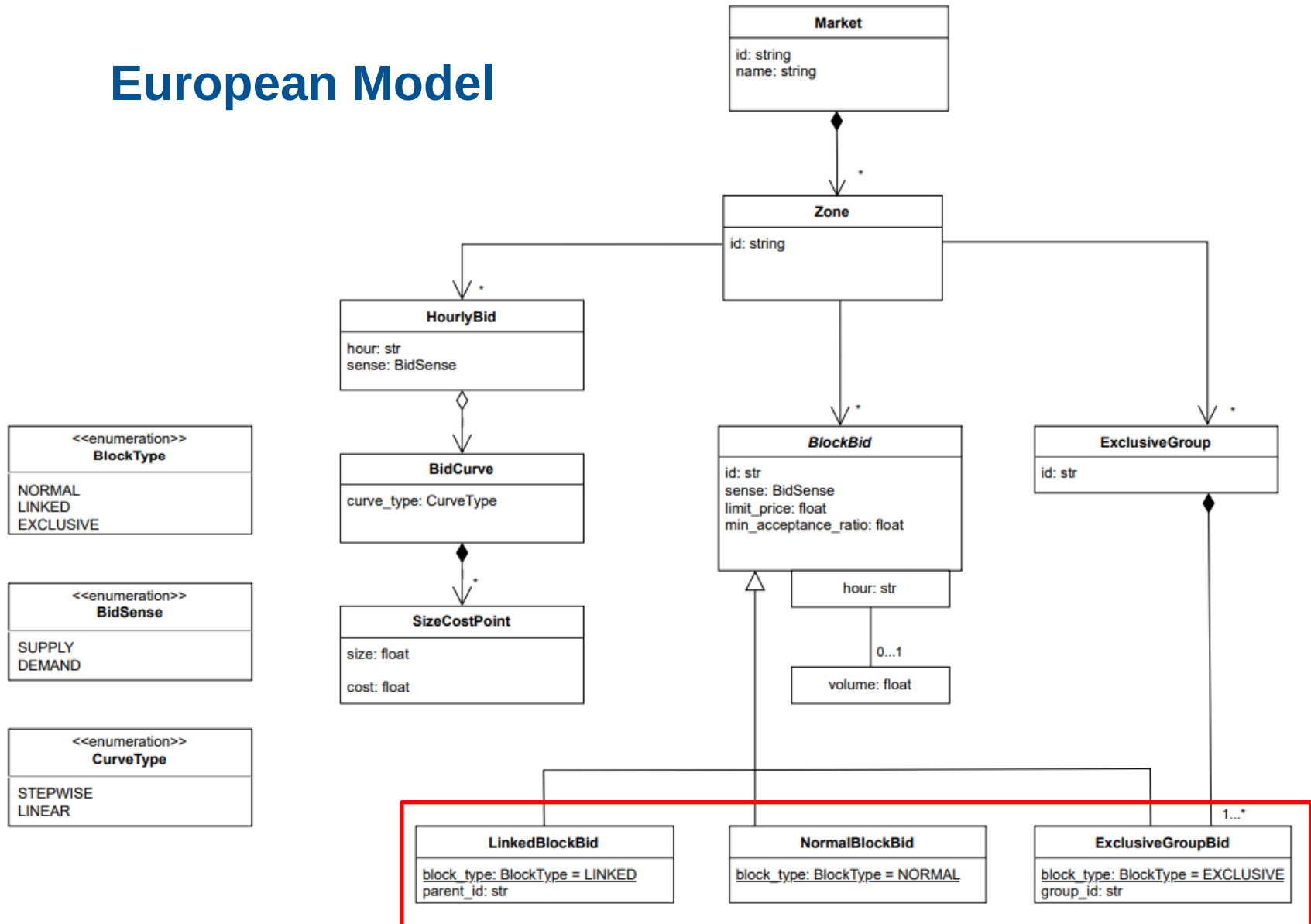
European Model



European Model

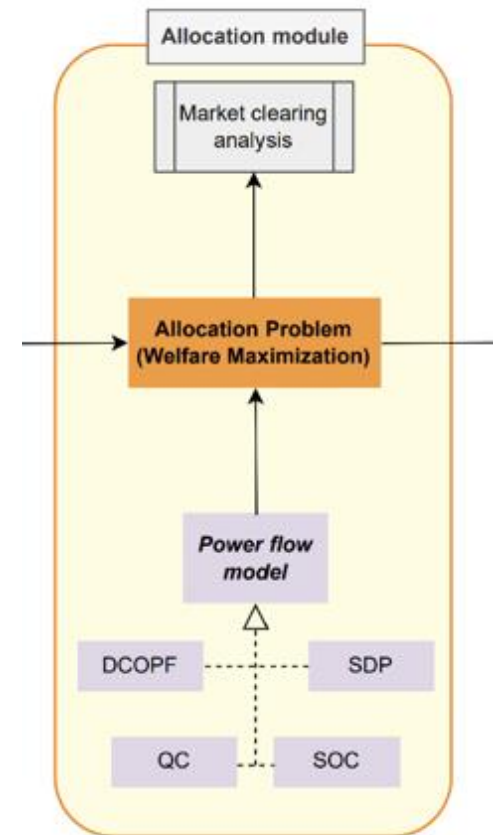


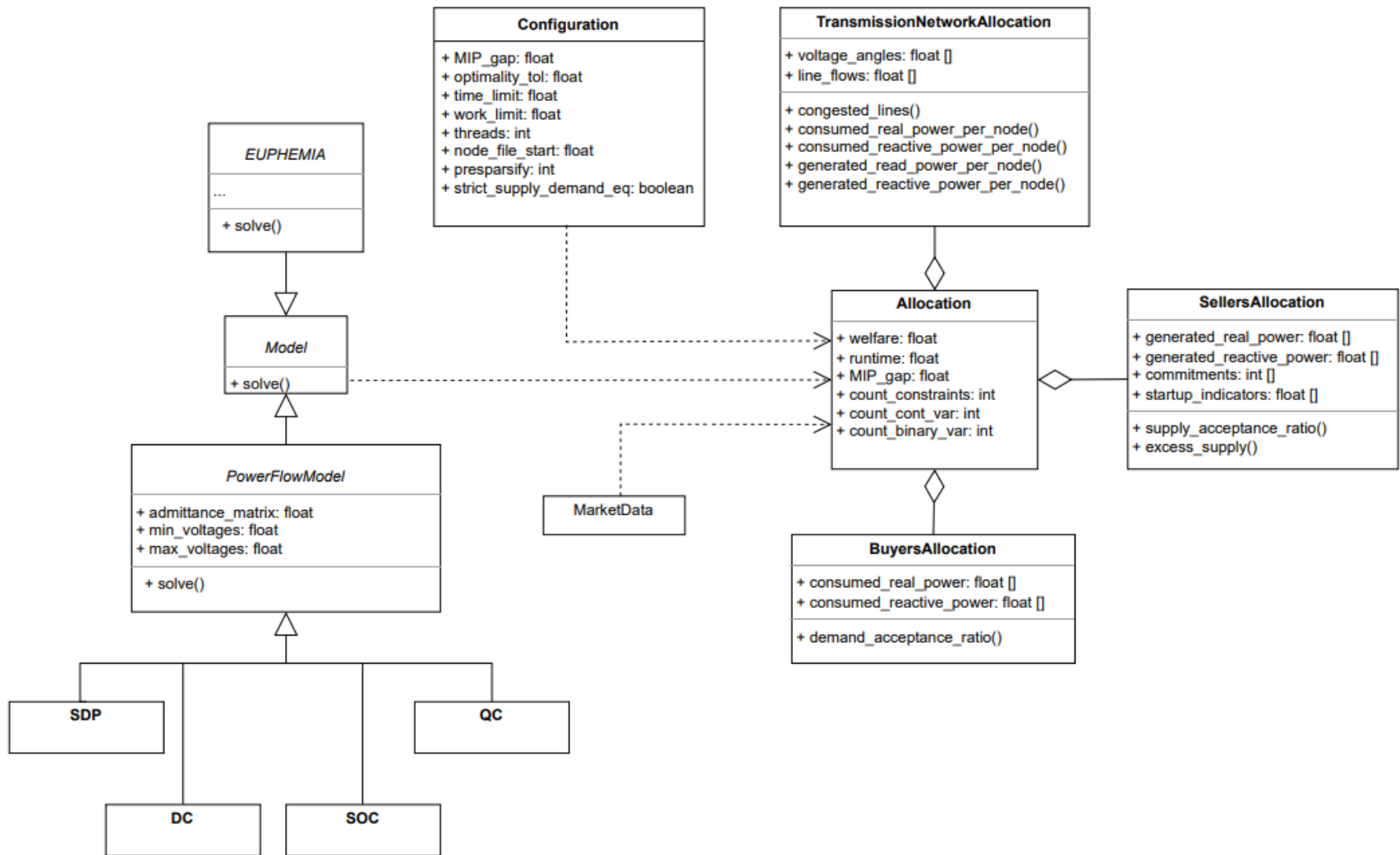
European Model

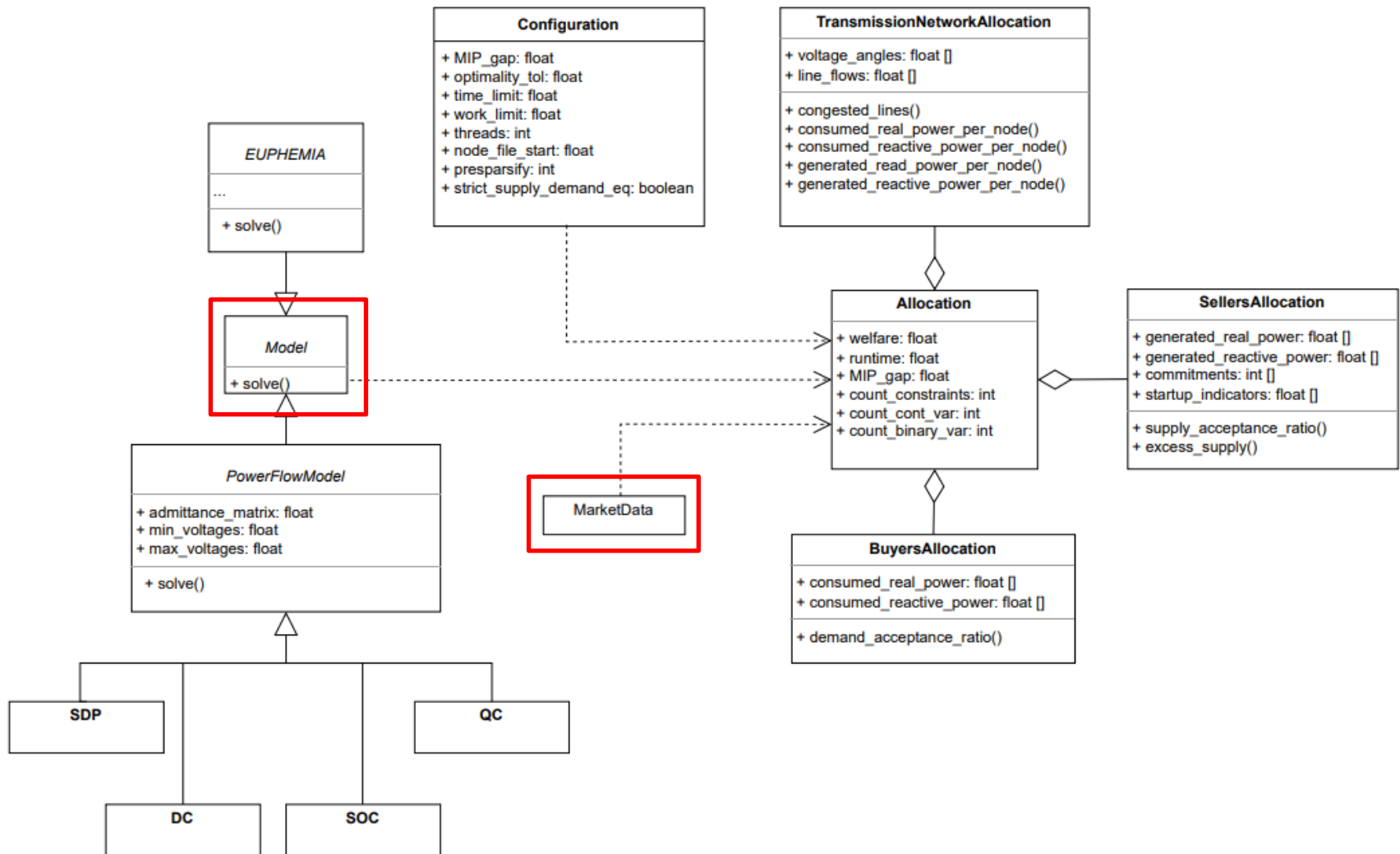


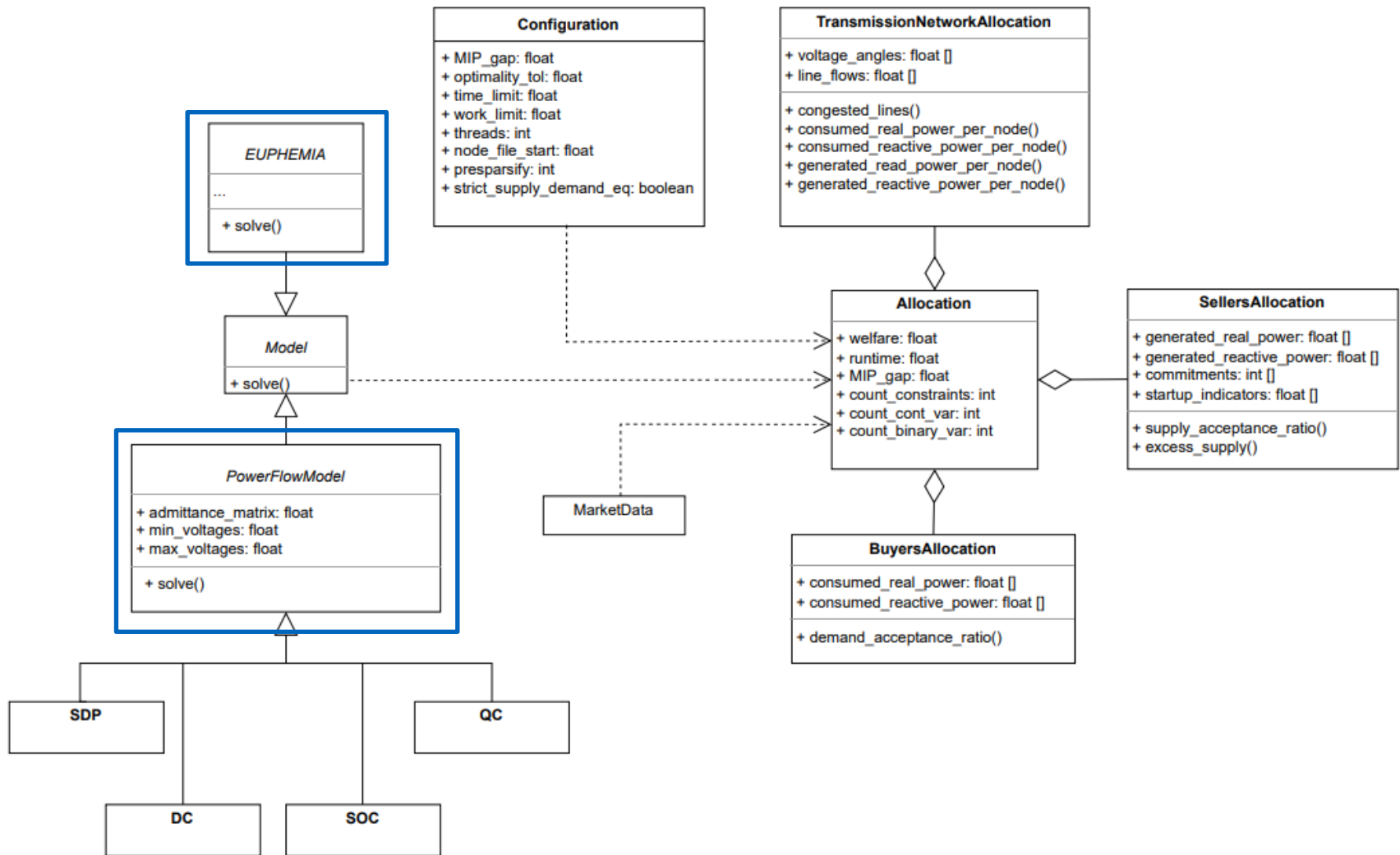
Allocation Module

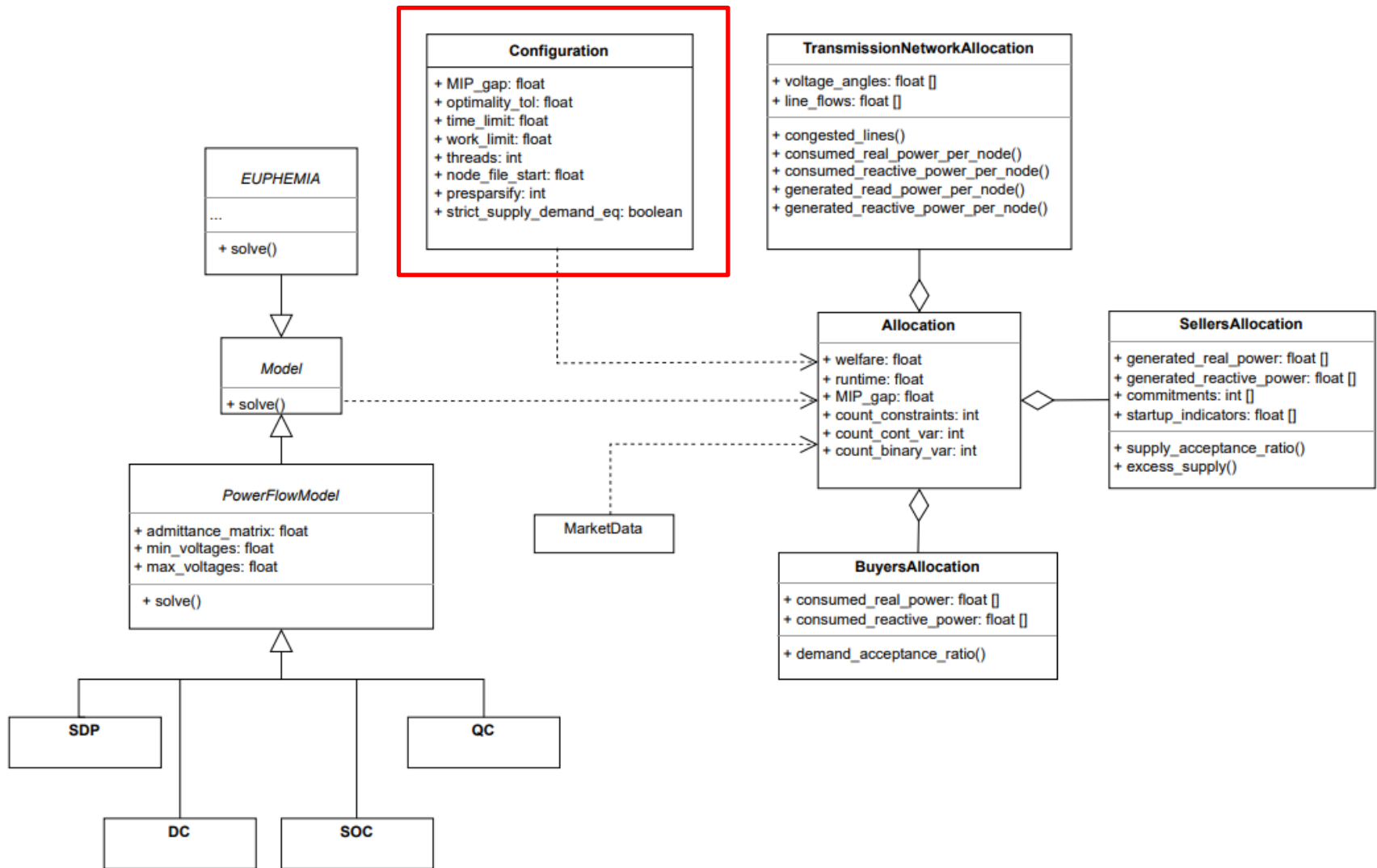
- Includes 2 submodules:
 1. Power Flow Models
 2. EUPHEMIA

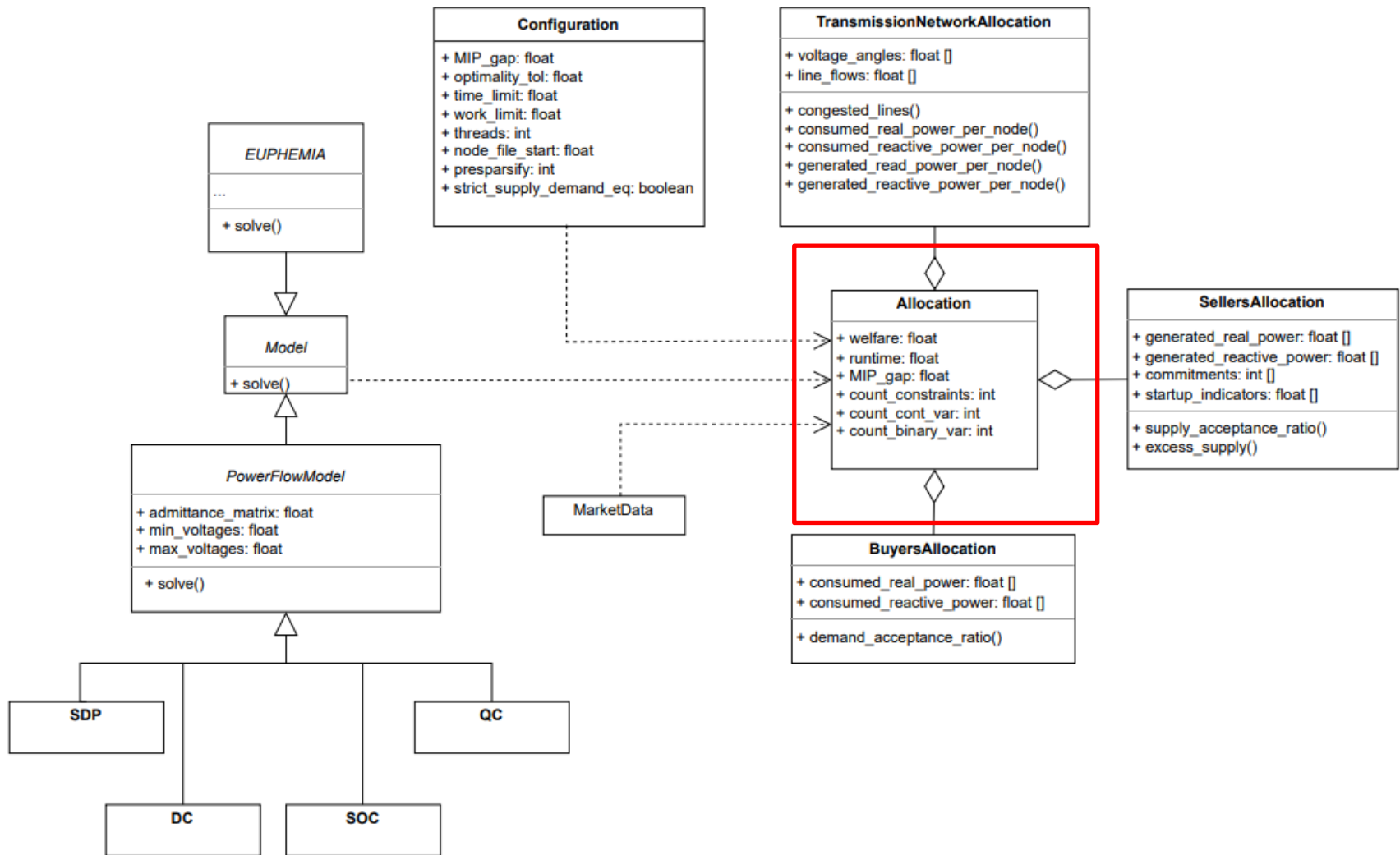


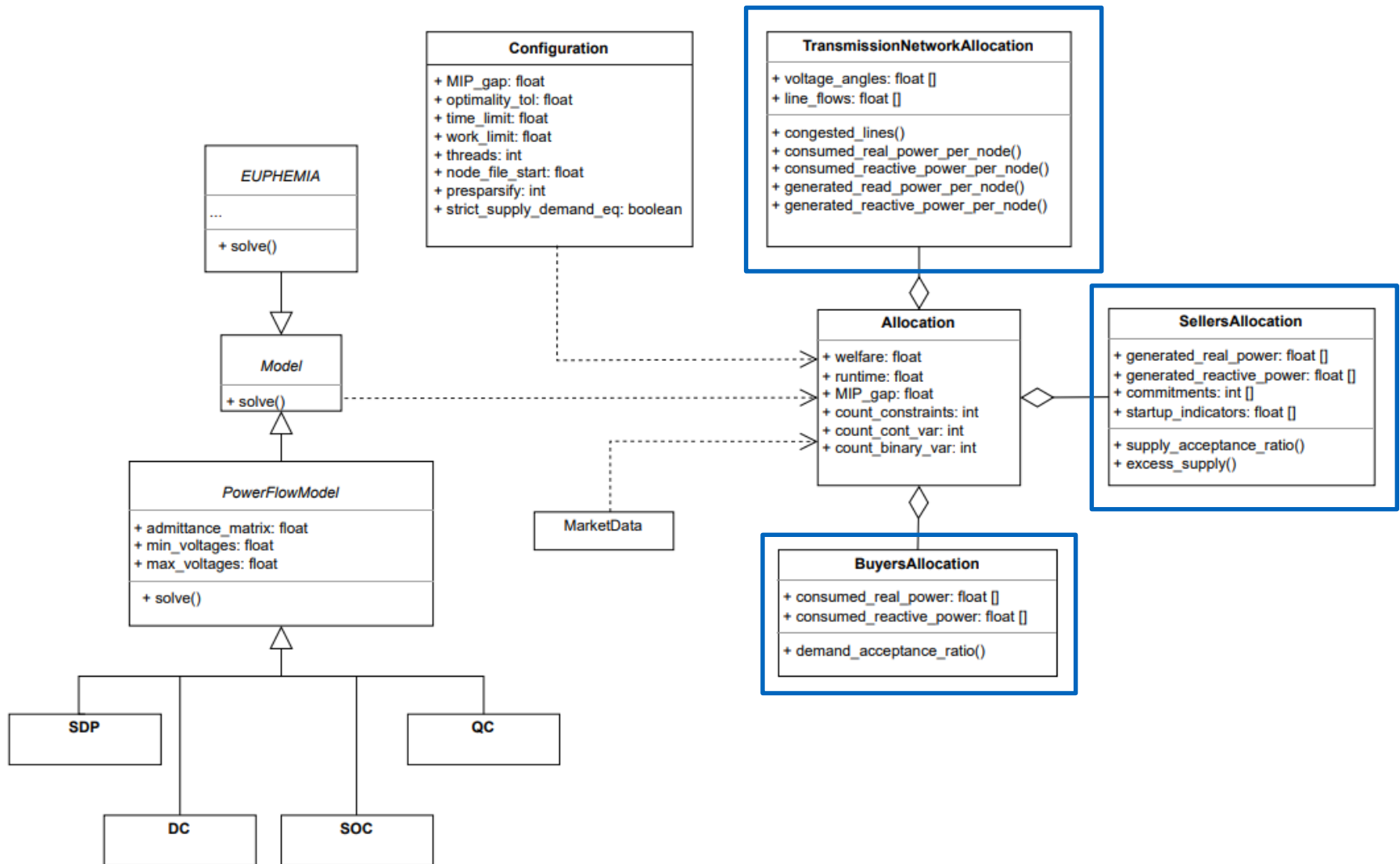






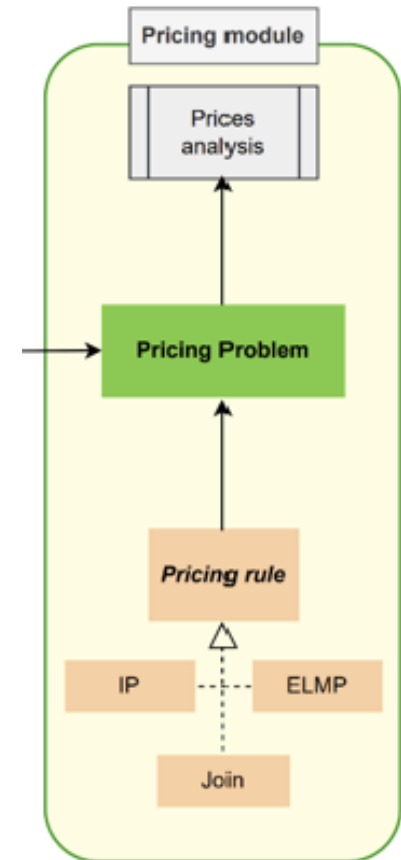


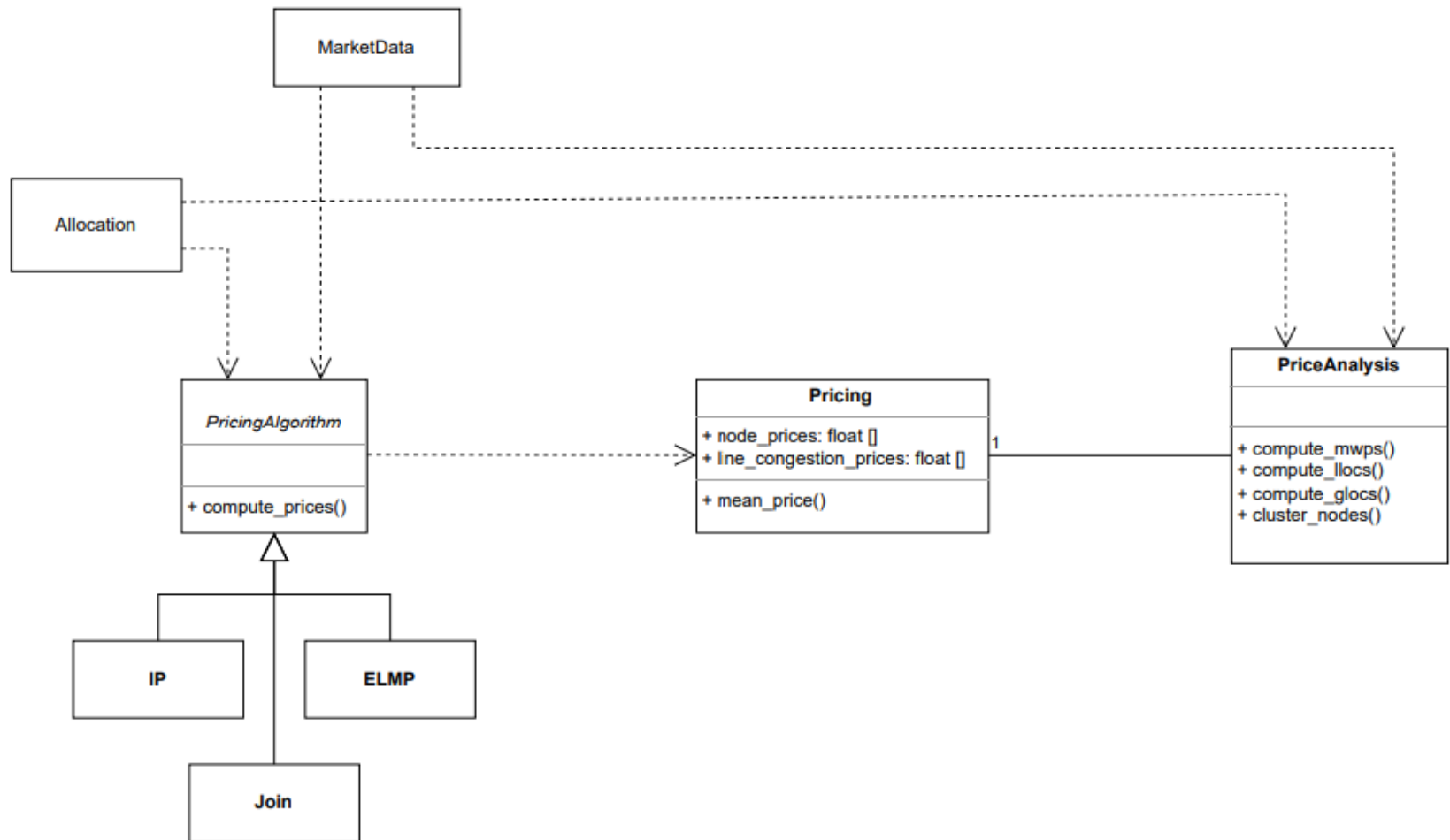


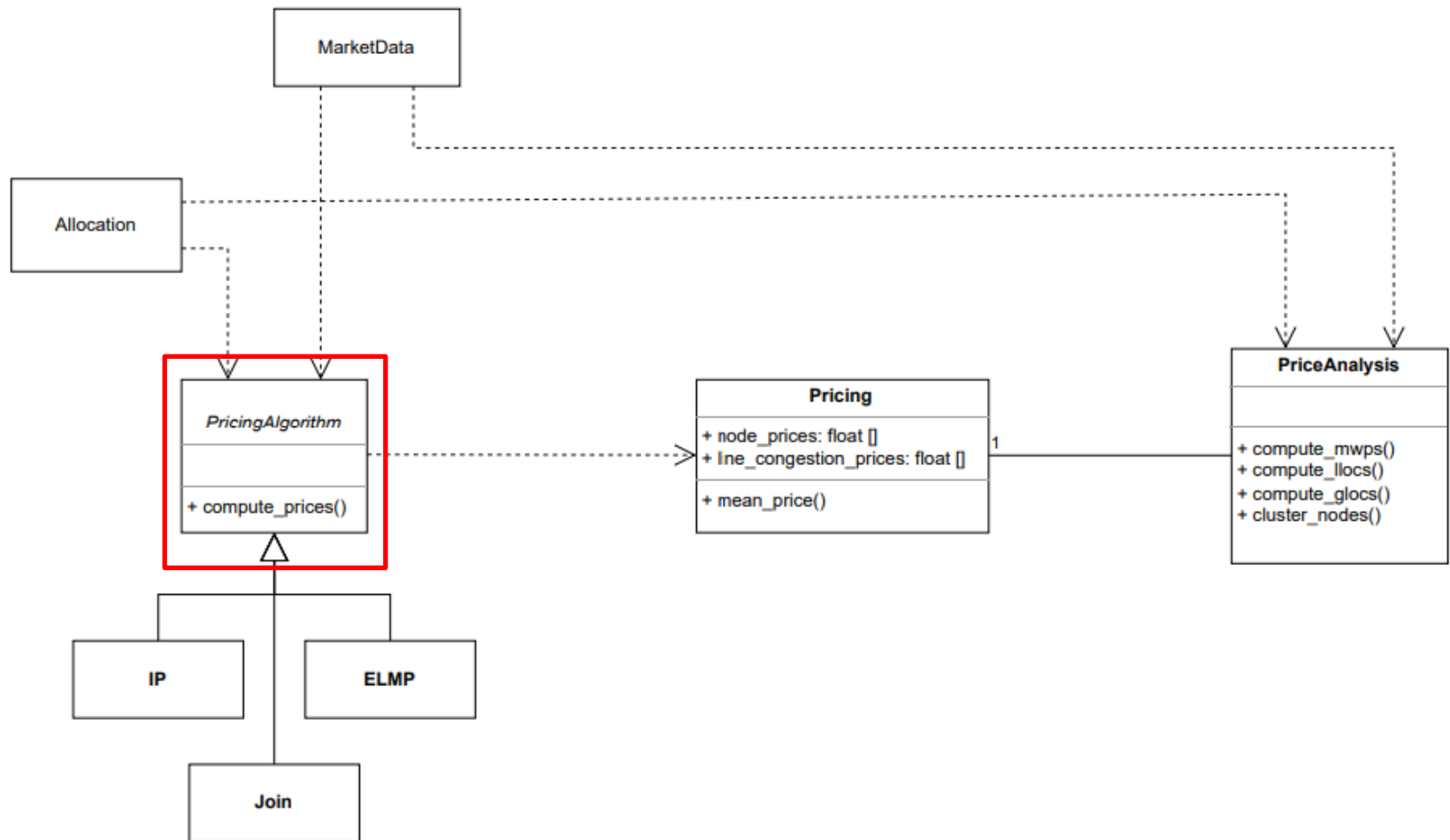


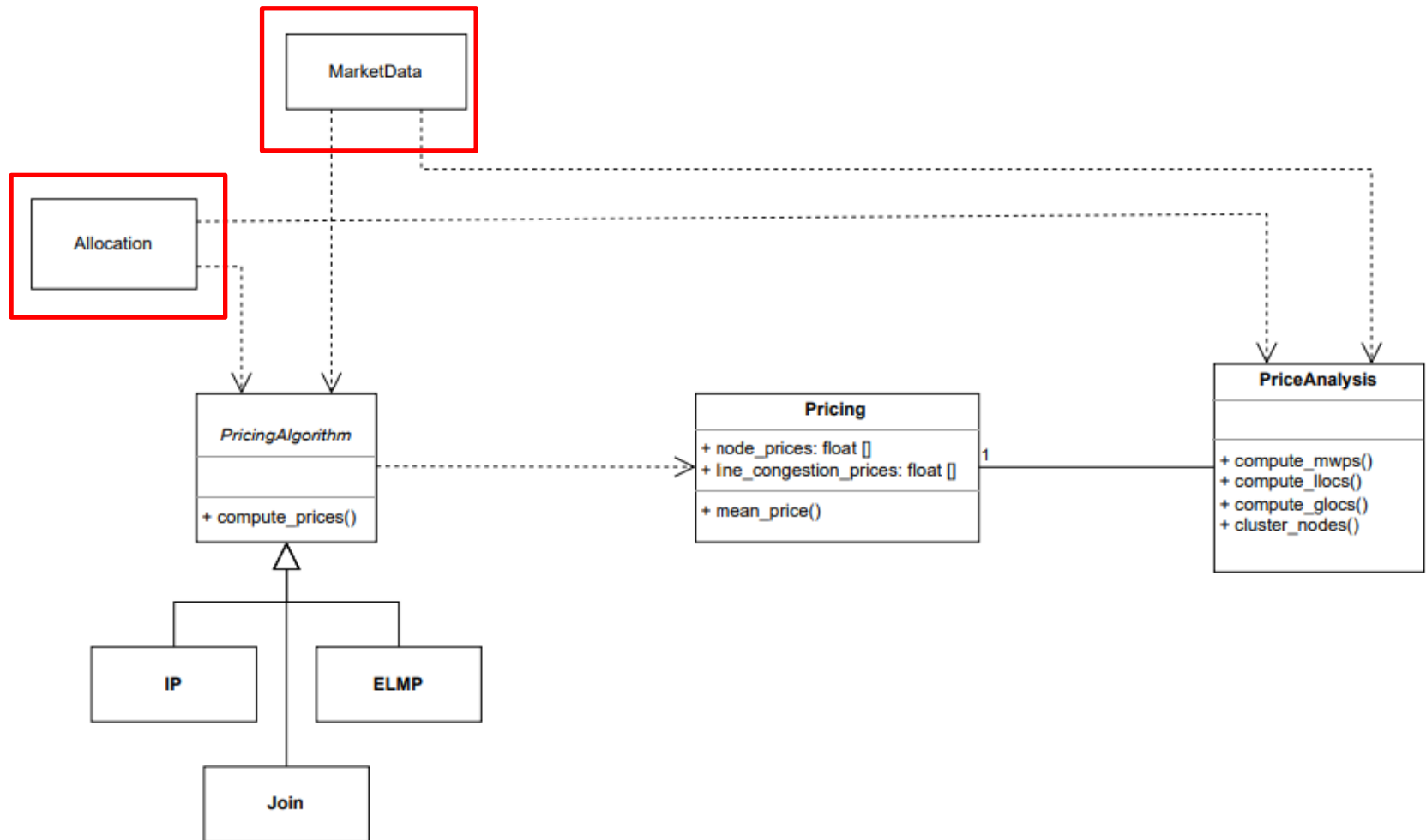
Pricing Module

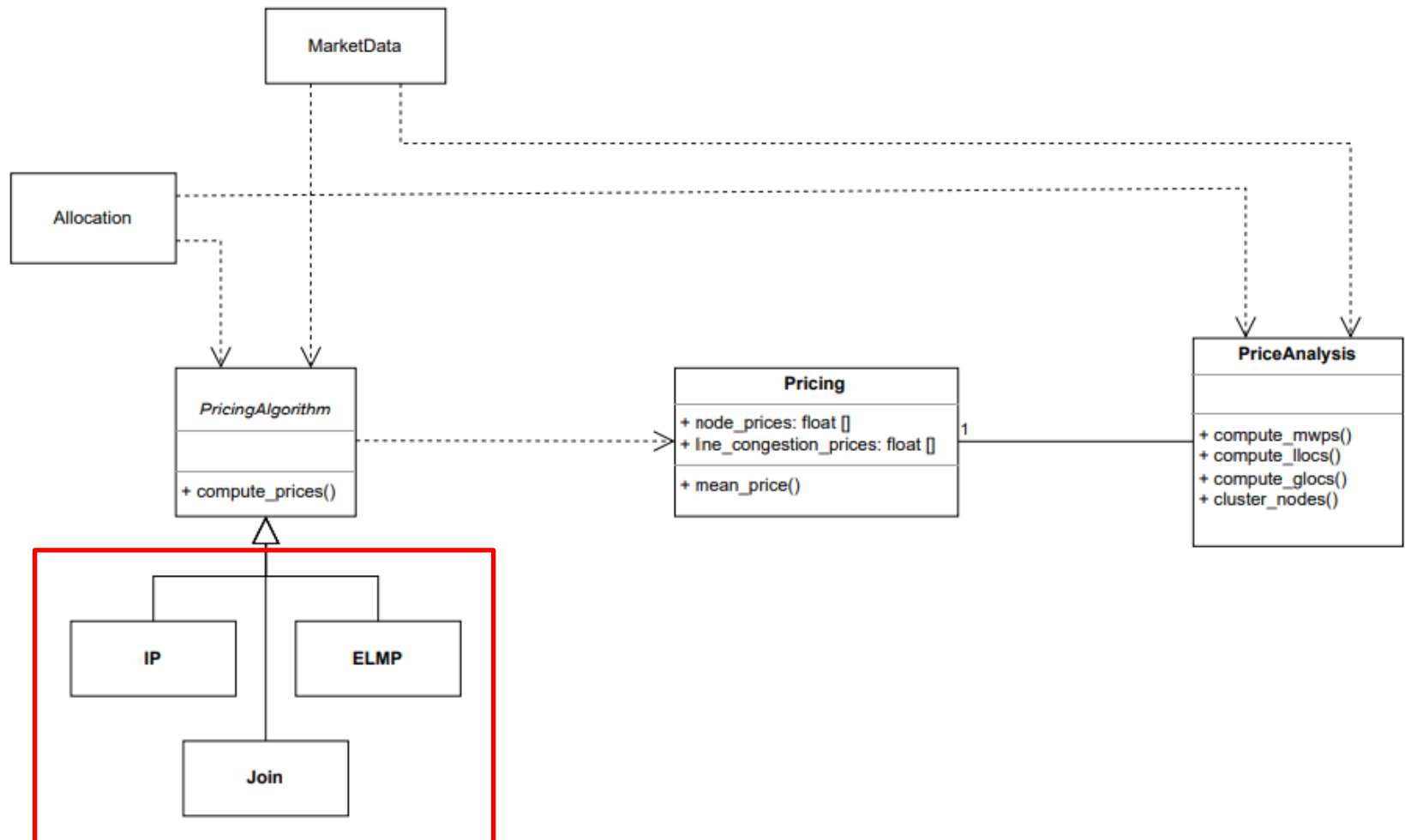
OOP → new pricing rules can be easily integrated

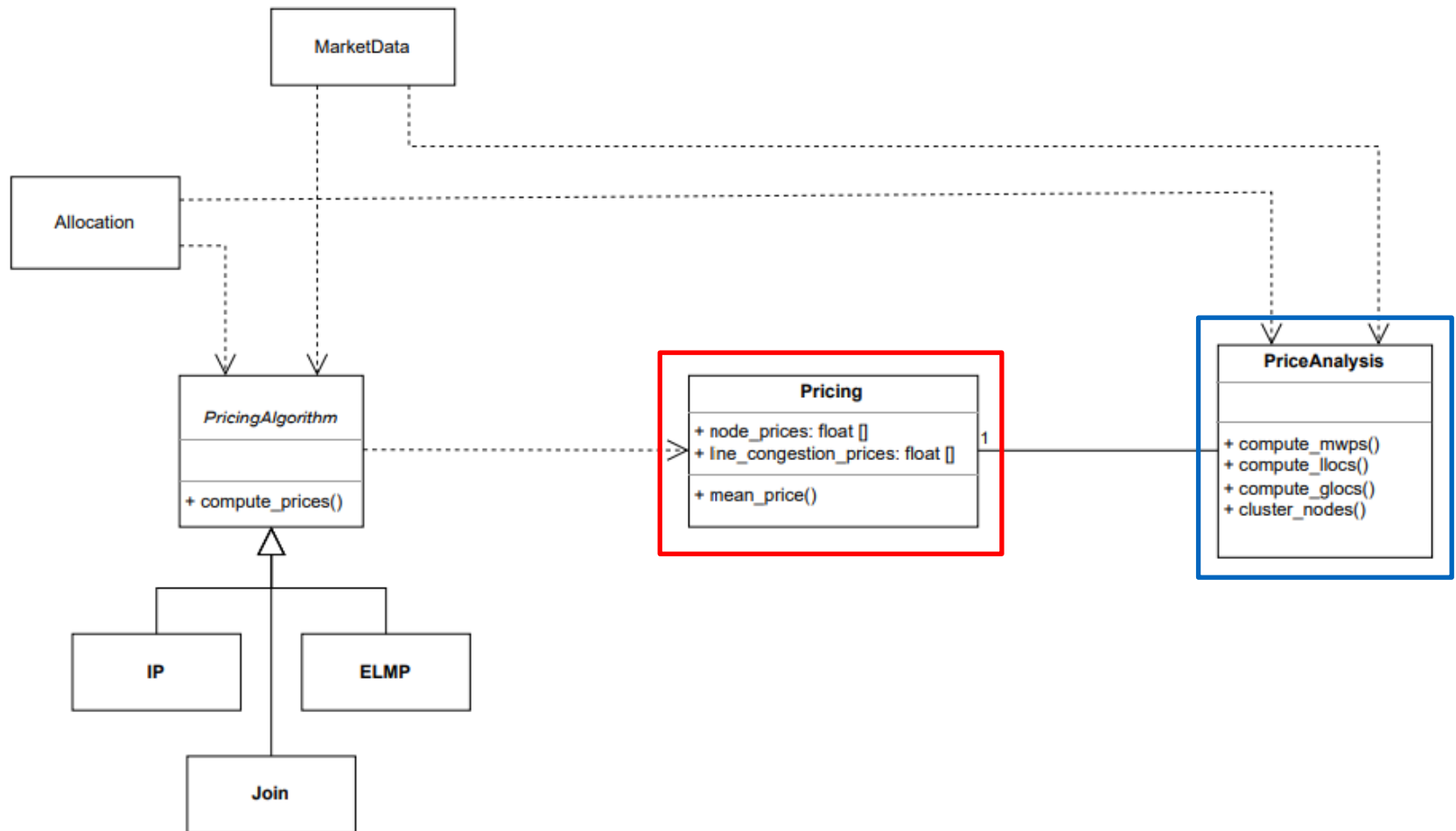












Demo

 tdaccount Add code demo. 3d19652 · 2 days ago	
 demo	Add code demo.
 framework_design	Add functional specification
 meetings/06.12.2023	Add slides from Mrs Rodriguez
 README.md	Add framework overview and meeting slides 06.12.2023
 framework_overview.png	Increase resolution of framework overview
 ganttt_diagram.png	Add framework overview and meeting slides 06.12.2023

Thanks!